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In [1]: import sqlite3

# Connect to SQLite database (creates file if it doesn't exist)
conn = sqlite3.connect("sales_data.db")
cursor = conn.cursor()

# Create sales table
cursor.execute("""
CREATE TABLE IF NOT EXISTS sales (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    product TEXT,
    quantity INTEGER,
    price REAL
)
""")

# Insert sample data
sales_data = [
    ("Laptop", 5, 700),
    ("Mobile", 10, 300),
    ("Tablet", 7, 250),
    ("Laptop", 3, 700),
    ("Mobile", 4, 300),
    ("Tablet", 2, 250)
]

cursor.executemany("INSERT INTO sales (product, quantity, price) VALUES (?, ?, ?)", sales_data)
conn.commit()
conn.close()
```

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In [3]: import pandas as pd
import sqlite3

# Connect to the database
conn = sqlite3.connect("sales_data.db")

# Query total quantity and revenue per product
query = """
SELECT
    product,
    SUM(quantity) AS total_qty,
    SUM(quantity * price) AS revenue
FROM sales
GROUP BY product
"""

# Load into pandas DataFrame
df = pd.read_sql_query(query, conn)

# Show the result
print("Sales Summary:")
print(df)

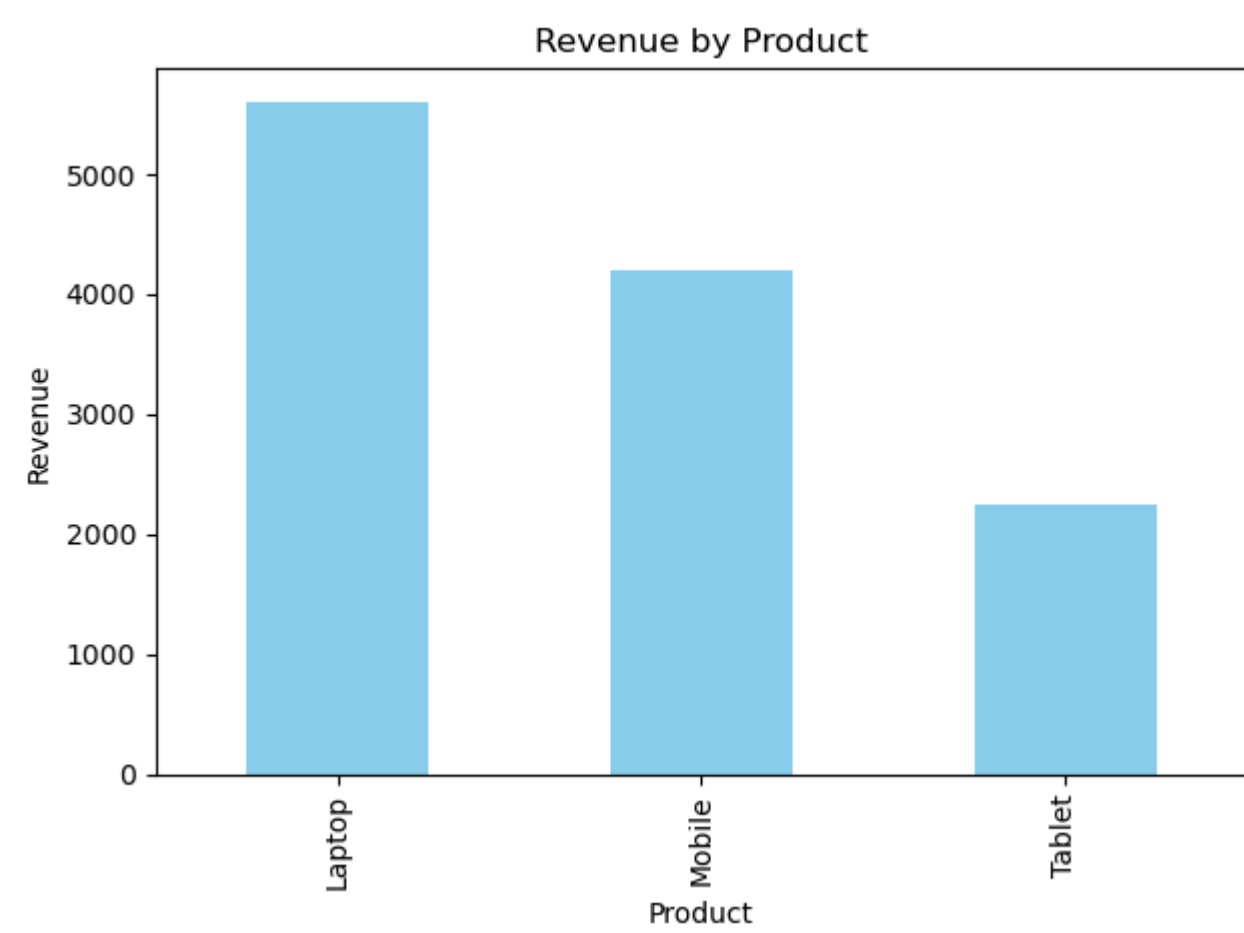
conn.close()
```

Sales Summary:

	product	total_qty	revenue
0	Laptop	8	5600.0
1	Mobile	14	4200.0
2	Tablet	9	2250.0

```
In [5]: import matplotlib.pyplot as plt

# Plot bar chart of revenue per product
df.plot(kind='bar', x='product', y='revenue', color='skyblue', legend=False)
plt.title("Revenue by Product")
plt.xlabel("Product")
plt.ylabel("Revenue")
plt.tight_layout()
plt.savefig("sales_chart.png") # Optional: saves the chart as a PNG
plt.show()
```



In []: